bargh-et-al-2015-automaticity-of-social-behavior--direct-e

Auto-assembled report. Section content is drawn from the summarization pipeline outputs.

Table of Contents

- Paper Synthesis
- Abstract
- Introduction (Paper-Level)
- Experiment 1 -- Introduction *(not available)*
- Experiment 1 -- Methods
- Experiment 1 -- Results and Discussion
- Experiment 2a and 2b -- Introduction *(not available)*
- Experiment 2a and 2b -- Methods
- Experiment 2a and 2b -- Results and Discussion
- Experiment 3 -- Introduction
- Experiment 3 -- Methods
- Experiment 3 -- Results and Discussion
- General Discussion
- Critique

Paper Synthesis

Paper:
bargh-et-al-2015-automaticity-of-social-behavior--direct-effects-of-trait-construct-and-stereotype-activation-on-a

Executive Summary

The paper investigates whether incidental priming of trait concepts or stereotypes can directly shape overt social behavior without conscious mediation. Across three experiments, priming rudeness accelerated participants' interruption of an experimenter, priming an elderly stereotype slowed walking speed, and priming African-American faces increased hostility ratings after a task error. These findings are presented as evidence that activated social constructs can automatically trigger behavior consistent with their content.

Reading Recommendation

Read Selectively

The paper offers novel cross-experiment evidence that trait and stereotype priming can directly affect overt behavior, which may be of theoretical interest. However, high-severity methodological issues (no power analysis, ambiguous outcome measures, and insufficient control of alternative mediational pathways) limit confidence in the claims. Scholars should focus on the Introduction, General Discussion, and the detailed methods/results of Experiment 1 (behavioral priming of rudeness) and Experiment 3 (subliminal stereotype-driven hostility), while skimming the less robust Experiment 2.

Background and Motivation

The authors identify a specific gap: although prior research has documented that trait concepts and stereotypes can be automatically activated and bias social perception, it remains unclear whether such passive priming effects extend to directly shaping overt behavioral responses. This gap is motivated by classic reviews of automaticity in social cognition (Bargh 1994) and foundational demonstrations that incidental activation influences interpretation of behavior (Higgins 1989; Srull & Wyer 1979). By targeting this unanswered question, the paper seeks to test whether priming of traits or stereotypes can produce measurable, non-conscious changes in actions such as interrupting, walking speed, and hostile reactions.

Theoretical Stance

Overall stance: Consistent Theoretical

The introduction posits that trait and stereotype representations can be automatically activated to produce behavior, and the general discussion repeatedly aligns with this claim, elaborating situationally linked behavior and related automaticity models without abandoning the original theoretical position.

Theories Discussed

- **automatic activation of behavioral responses** *(focal)*: The paper's central claim that trait concepts and stereotypes, once mentally represented, can become automatically activated and directly drive overt social behavior.
- **situationally linked behavior** *(focal)*: A refinement proposed in the discussion that behavioral responses are linked to specific situational cues, allowing automatic priming effects when the situation matches the activated representation.
- **automatic perception-behavior link** *(alternative)*: A theoretical perspective (e.g., Berkowitz 1984; Carver et al. 1983) suggesting that perception and behavior are

automatically coupled, offered as a compatible but distinct account of the findings.

- **behavioral schema model** *(background)*: Carver et al.'s (1983) model of behavioral schemas that organize stimulus-response relations, cited to situate the present results within broader automaticity literature.
- **mediational model of behavior (perception-mediated)** *(alternative)*: The competing account that priming influences behavior indirectly via altered social perceptions or judgments, briefly acknowledged and argued against.
- **automatic stereotype-based self-fulfilling prophecy** *(background)*: Snyder, Tanke & Berscheid (1977) notion that stereotypes can produce self-fulfilling effects without conscious mediation, referenced in the discussion.
- **automatic empathic reactions** *(background)*: A descriptive account of how perception-behavior links may underlie empathic synchrony between interaction partners.
- **general automaticity of mental representations** *(background)*: Foundational work (Bargh 1994; Higgins 1989; Wyer & Srull 1989) on incidental activation of traits, attitudes, and stereotypes, providing the broader theoretical context.

Study Design

- **Number of experiments:** 3
- **Predictors examined:** priming (rude vs. polite vs. neutral), elderly stereotype prime vs. neutral, African American face prime vs. Caucasian face prime
- **Outcomes examined:** interruption latency (seconds until participant interrupts experimenter), walking time (seconds to walk 9.75 m corridor), hostility rating (5-point unipolar scale averaged across coders and experimenter)
- **Materials:** All experiments used a scrambled-sentence task to deliver the prime (trait-related adjectives for rude/polite/neutral; elderly-related words; or subliminal face photographs). Experiment 1 measured interruption latency with a stopwatch; Experiment 2 measured walking speed with a hidden stopwatch in a 9.75 m hallway; Experiment 3 recorded facial reactions after a computer-error and coded hostility from video by blind coders.
- **Design types:**
- exp_1: between-subjects 3-level (rude, polite, neutral) design
- exp_2: 2 × 2 between-subjects design (elderly vs. neutral prime × experiment 2a vs. 2b)
- exp_3: between-subjects 2-level (African American vs. Caucasian face) design

Findings Summary

Across the three experiments, priming of a trait or stereotype consistently produced behavior aligned with the activated construct: rude priming accelerated participants' interruption of an experimenter, elderly-stereotype priming slowed walking speed, and African-American-face priming increased hostility ratings after a computer error. The effects were observed in distinct behavioral domains (verbal interruption, locomotion, and facial hostility) and persisted despite the lack of conscious awareness of the primes. However, the experiments diverge in the specificity of the behavioral measures--interruption latency may reflect general assertiveness rather than rudeness, walking speed could be influenced by mood despite authors' claims, and the hostility rating relies on indirect facial expressions--highlighting potential construct mismatches in two of the studies.

Evidence-Claim Alignment

Supported Claims

- Priming a rude trait concept leads to faster interruption of the experimenter (Exp 1): supported by a significant main effect ($F(2,33)=5.76$, $p=.008$) and planned t-tests ($ps<.04$) showing the rude-prime mean latency shorter than polite and neutral.
- Elderly stereotype priming slows walking speed (Exp 2a and 2b): supported by significant t-tests ($t(28)=2.86$, $p<.01$; $t(28)=2.16$, $p<.05$) with slower times in the elderly-prime condition.
- African-American face priming increases hostility ratings after a computer error (Exp 3): supported by a main effect in MANOVA ($F(1,39)=6.95$, $p<.05$) with higher mean hostility ($M=2.79$) versus Caucasian prime ($M=2.13$).

Unsupported or Weakly Supported Claims

- Interruption latency reflects a direct manifestation of rudeness rather than general assertiveness: the measure captures willingness to intervene, and the critique notes a mismatch between the theoretical construct (rude behavior) and the operationalization, leaving the claim weakly supported.
- Elderly-prime effect on walking speed is nonconscious and mood-independent: although mood differences were non-significant in MANOVA, descriptive data show a more positive mood in the elderly-prime group, and the authors claim mood does not mediate the effect without mediation analysis, making the claim only weakly supported.
- Hostility ratings after African-American priming directly reflect stereotype-driven hostility: the behavioral measure (hostility ratings of facial expressions after an error) is indirect, and

the authors do not justify that this captures the hypothesized stereotype-based hostile response, rendering the claim weakly supported.

Contradictions or Tensions

- Experiment 1: Authors assert a direct, preconscious effect of rude priming on interruption behavior, yet they acknowledge an alternative perceptual explanation (participants may have perceived the experimenter differently) and provide no manipulation checks, creating tension between the claim and the evidence.
- Experiment 2: The claim of a nonconscious stereotype effect conflicts with reported mood differences (more positive mood in elderly-prime participants) and the lack of statistical mediation testing, producing a tension between the asserted mood-independence and the data.
- General discussion vs. evidence: The authors argue that behavior can be triggered directly without mediation, but across all experiments they do not empirically rule out mediational pathways (e.g., perception, mood), leading to an internal inconsistency between the theoretical stance and the presented evidence.

Strengths

- Demonstrates converging evidence across three distinct behavioral domains (verbal interruption, locomotion speed, and facial hostility) that priming of trait concepts or stereotypes can influence overt actions.
- Employs a consistent priming technique (scrambled-sentence task) across experiments, enhancing methodological coherence and allowing cross-experiment comparisons.
- Provides clear statistical support for the primary hypotheses in each experiment (significant main effects of priming on interruption latency, walking speed, and hostility ratings).
- Integrates theoretical perspectives on automaticity (situationally linked behavior, perception-behavior link, behavioral schema model) to contextualize findings within broader social cognition literature.
- Highlights the potential for non-conscious, direct behavioral effects, challenging mediation-based models and suggesting novel mechanisms for self-fulfilling prophecy processes.

Weaknesses and Concerns

- Key outcome measures often mismatch the targeted constructs: interruption latency may index general assertiveness rather than rudeness, walking speed could be mood-driven despite claims of non-conscious effects, and hostility ratings rely on indirect facial expressions, weakening the link between priming and the theorized behavior.
- Alternative mediational pathways (e.g., perceptual reinterpretation of the experimenter, mood effects) are not empirically ruled out; no manipulation checks or mediation analyses are provided, leaving the asserted direct, preconscious effects insufficiently substantiated.
- Statistical reporting is incomplete: effect sizes and confidence intervals are omitted, multiple comparisons lack correction, and the 2 × 2 design in Experiment 2 is not fully analyzed (missing main-effect and interaction tests).
- Methodological transparency is limited: no a priori power analyses, undefined exclusion criteria, absent counterbalancing details, and missing reliability estimates for coded dependent variables raise concerns about statistical power and measurement precision.
- The paper does not address the potential influence of participant characteristics (e.g., individual differences in behavioral repertoires or prejudice) despite discussing self-fulfilling prophecy mechanisms, which may limit the generalizability of the findings.

Key References

- **Bargh (1994)** -- foundational theory
- **Higgins (1989)** -- foundational theory
- **Wyer & Srull (1979)** -- foundational theory
- **Berkowitz (1984)** -- competing theoretical account
- **Carver et al. (1983)** -- competing theoretical account
- **Snyder, Tanke, & Berscheid (1977)** -- key empirical precedent
- **Devine (1989)** -- key empirical precedent
- **Bargh et al. (1986)** -- primary methodological precedent
- **Fazio, Jackson, Dunton, and Williams (1995)** -- key empirical precedent
- **Bargh (1989)** -- primary methodological precedent
- **Wegner (1994)** -- primary methodological precedent
- **Brewer (1988)** -- foundational theory
- **Devine (1989)** -- foundational theory
- **Fiske & Neuberg (1990)** -- foundational theory
- **Hayes-Roth (1977)** -- foundational theory

Abstract

Preserved from source without summarization.

Previous research has shown that trait concepts and stereotypes become active automatically in the presence of relevant behavior or stereotyped-group features. Through the use of the same priming procedures as in previous impression formation research, Experiment 1 showed that participants whose concept of rudeness was primed interrupted the experimenter more quickly and frequently than did participants primed with polite-related stimuli. In Experiment 2, participants for whom an elderly stereotype was primed walked more slowly down the hallway when leaving the experiment than did control participants, consistent with the content of that stereotype. In Experiment 3, participants for whom the African American stereotype was primed subliminally reacted with more hostility to a vexatious request of the experimenter. Implications of this automatic behavior priming effect for self-fulfilling prophecies are discussed, as is whether social behavior is necessarily mediated by conscious choice processes.

Introduction (Paper-Level)

Section: paper-level

Research Gap / Motivation

It remains unclear whether the passive, automatic effects of priming extend beyond social perception to directly influence behavioral responses.

Theoretical Framework

automatic activation of behavior

> But assuming that behavioral responses to situations are also represented mentally, as are stereotypes and attitudes, they should also be capable of becoming automatically activated, by the same principles that govern the development of automaticity of other representations.

Mechanistic Claims

mental representation of behavioral responses

> But assuming that behavioral responses to situations are also represented mentally, as are stereotypes and attitudes, they should also be capable of becoming automatically activated, by the same principles that govern the development of automaticity of other

> representations.

Constructs of Interest

- **priming of trait constructs or stereotypes** *(predictor)*: the incidental activation of trait concepts or stereotypes by the current situational context
- **behavioral responses** *(outcome)*: actions or reactions to situations that are represented mentally and can be automatically activated

Rationale for Predictor

Priming is argued to activate knowledge structures automatically, providing a mechanism by which prior exposure can influence later cognition and behavior.

Rationale for Outcome

If behavioral responses are mentally represented like stereotypes and attitudes, they can be automatically activated by the same principles governing other automatic representations.

Key Terminology

priming: the incidental activation of knowledge structures, such as trait concepts and stereotypes, by the current situational context

> Priming refers to the incidental activation of knowledge structures, such as trait concepts and stereotypes, by the current situational context.

Background Research

- Bargh (1994) reviewed evidence that priming of trait constructs and stereotypes influences subsequent impressions of others.
- Higgins (1989) documented that incidental activation of knowledge structures can have unintended effects on interpretation of behavior.
- Wyer & Srull (1989) highlighted the passive influence of prior use of trait constructs on later perception.

Missing or Unclear

- specific hypotheses
- overview of studies

- alternative theories

All Citations Referenced

- **Bargh (1994)** -- review of priming effects on social perception and automaticity
- **Higgins (1989)** -- effects of incidental activation on interpretation of behavior
- **Wyer & Srull (1989)** -- passive influence of prior trait use on perception

Additional Quoted Evidence

> Priming refers to the incidental activation of knowledge structures, such as trait concepts and stereotypes, by the current situational context.

> Many studies have shown that the recent use of a trait construct or stereotype, even in an earlier or unrelated situation, carries over for a time to exert an unintended, passive influence on the interpretation of behavior.

> But assuming that behavioral responses to situations are also represented mentally, as are stereotypes and attitudes, they should also be capable of becoming automatically activated, by the same principles that govern the development of automaticity of other representations.

Experiment 1 -- Introduction

Not available for this paper.

Experiment 1 -- Methods

Section: exp_1

Participants

- **N:** 34 students at New York University
- **Recruitment:** Introductory Psychology course at NYU
- **Testing context:** Individual participants tested one at a time in a laboratory setting

Design

Design statement: 3 (priming: rude vs. polite vs. neutral) between-subjects design.

Predictors:

- **priming**
- Levels: rude, polite, neutral
- Manipulation: between-subjects
- Factor type: nominal

Counterbalancing: Participants were randomly assigned to one of the three priming conditions with approximately equal numbers per condition.

Outcome Measures

- **interruption latency:** Time (in seconds) from when the participant entered the hallway until they spoke to the experimenter, measured with a stopwatch; maximum of 10 minutes.

Materials

- **Description:** Scrambled Sentence Test: participants construct a grammatically correct four-word sentence from five scrambled words as quickly as possible.
- **Source:** Adapted from Srull & Wyer (1979)
- **Stimulus pool size:** 30 items per test version
- **Stimuli presented per participant:** 30 items per participant
- **Additional details:**
- Three versions (rude, polite, neutral) differing in 15 critical adjectives/verbs
- Items presented in scrambled order

Procedure

Ordered steps:

1. Participant arrives at waiting area and is greeted by experimenter.
2. Consent obtained; participant receives envelope with one scrambled-sentence test.
3. Participant completes the 30-item scrambled-sentence test (≈5 min).
4. Participant exits to hallway and approaches second room where experimenter and confederate are conversing.
5. Experimenter gives subtle prearranged sign; confederate starts stopwatch.
6. Participant waits; stopwatch records time until participant speaks (interrupts).

7. If no interruption within 10 min, stopwatch stops.
8. Participant completes brief anagram puzzle (~2 min).
9. Partial debrief and questionnaire about perceived influence of the first task.
10. Full debrief after completing survey.

- **Experimenter presence:** present in the laboratory rooms
- **Presentation format:** not reported
- **Stimulus timing:** self-paced sentence construction; interruption latency measured with stopwatch, max 10 min

Preregistration and Analysis Plan

- **Preregistration:** not reported

Missing or Unclear

- Compensation details not provided
- Participant demographics (age, gender, etc.) not reported
- Exclusion criteria not stated
- Power analysis not reported
- Apparatus/hardware/software details not provided
- Exact stimulus timing for scrambled-sentence test not specified beyond self-paced
- Total session duration not reported
- Data analysis plan not described
- Preregistration status not indicated

Citations Referenced in This Section

- **Srull & Wyer (1979)** -- source of the Scrambled Sentence Test used as the priming manipulation

Quoted Evidence from Source

> A total of 34 students at New York University who were enrolled in the Introductory Psychology course participated in the experiment...

> Three versions of the scrambled-sentence test were constructed: ... rude, polite, or neutral priming condition.

> Our dependent measure was the amount of time the participant would wait until interrupting the conversation...

Experiment 1 -- Results and Discussion

Section: exp_1

Statistical Approach

A one-way analysis of variance (ANOVA) of these data, with priming condition as the single factor, revealed a significant main effect. Within the significant main effect, simple t tests revealed that the rude prime condition mean was significantly shorter than each of the other two means. Because we made a specific directional prediction regarding the ordering of the three percentages, rather than an omnibus prediction of any difference among the three, we applied the test for a linear trend in proportions. To assess whether the priming manipulations had resulted in differential perceptions of the experimenter's politeness, we examined the ratings participants made on the "Experimental Participation Survey."

Key Findings

Participants in the rude priming condition interrupted significantly faster than those in the neutral or polite conditions.

- **Test:** one-way ANOVA
- **Effect type:** main effect
- **Variables in model:** priming condition, time to interruption

Verbatim statistical report:

> $F(2,33) = 5.76$, $p = .008$

Verbatim descriptives:

> Participants in the rude priming condition interrupted significantly faster ($M = 326$ s) than did participants in the neutral ($M = 519$ s) or polite ($M = 558$ s) priming conditions.

The rude prime condition mean was significantly shorter than each of the other two means.

- **Test:** simple t tests
- **Effect type:** planned comparison
- **Variables in model:** priming condition, time to interruption

Verbatim statistical report:

> both ps < .04

The proportion of participants who interrupted the experimenter increased as a function of the trait concept primed, from politeness through no priming to rudeness.

- **Test:** test for a linear trend in proportions
- **Effect type:** linear trend
- **Variables in model:** priming condition, percentage of participants who interrupted

Verbatim statistical report:

> Z = 2.32, p = .02, two-tailed

- Citations: Snedecor & Cochran (1980)

There was no reliable difference in the ratings of the experimenter's politeness among the three priming conditions.

- **Test:** one-way ANOVA
- **Effect type:** main effect
- **Variables in model:** priming condition, experimenter's politeness ratings

Verbatim statistical report:

> F(2,28) < 1

Verbatim descriptives:

> polite prime condition: M = 0.6, rude prime condition: M = 0.4, neutral prime condition: M = 0.8

Authors' Interpretations

direct effect on behavior

> Whereas our interpretation of this result is that the priming manipulation produced a direct, preconscious effect on the participants'subsequent behavior in a social interaction, an alternative explanation in terms of the social perceptual effects of priming must be considered.

- Citations: Higgins & King (1981)

automatic activation of trait concepts

> The results of Experiment 1 showed that the passive, automatic activation of a trait concept results in traitlike behavior by the individual, but there are two ways in which trait concepts can be activated directly by the environment.

- Citations: Newman & Uleman (1989); Srull & Wyer (1979)

stereotype activation

> Another way in which a given trait concept can be activated automatically, however, is by its membership in a larger schema, such as a stereotype. Stereotypes of social groups consist, in part, of constellations of interrelated trait concepts (e.g., Brewer,1988; Devine,1989; Fiske & Neuberg,1990) that become active in an all-or-none fashion (see Hayes-Roth,1977) in the presence of the features of a group member.

- Citations: Brewer (1988); Devine (1989); Fiske & Neuberg (1990); Hayes-Roth (1977)

priming effects on behavior

> The present priming effect on behavior is much more substantial than the typical priming effect in impression formation research (cf. Bargh et al.,1986; Srull & Wyer,1979). The powerful effects of priming on behavioral relative to perceptual dependent variables is a topic to which we return in the General Discussion.

- Citations: Bargh et al. (1986); Srull & Wyer (1979)

Authors' Stated Links to Their Own Predictions

linear trend in proportions

> Because we made a specific directional prediction regarding the ordering of the three percentages, rather than an omnibus prediction of any difference among the three, we applied the test for a linear trend in proportions (Snedecor & Cochran,1980, pp.206-208). The specific hypothesis tested was that the proportion of participants who interrupted the experimenter would increase as a function of the trait concept primed, from politeness through no priming to rudeness. In support of the hypothesis, the test revealed a reliable linear trend ($Z = 2.32$, $p = .02$, two-tailed).

- Citations: Snedecor & Cochran (1980)

Alternative Interpretations Discussed by Authors

- This alternative centers on how the participant may have interpreted or perceived the experimenter's behavior during his conversation with the confederate. If the experimenter's behavior was ambiguous enough to permit different impressions of him, depending on the relative accessibility of the participant's relevant trait constructs, it is possible that rude-primed participants were more likely to perceive the experimenter as rude because he was ignoring them and attending to the confederate, whereas polite-primed participants were more likely to perceive the experimenter as polite because he so patiently dealt with the confederate's questions. The participants' own behavior to the experimenter (i.e., whether they interrupted him) might have been based on these differential perceptions (as has been demonstrated in studies by Neuberg,1988, and Herr,1986) instead of as a direct effect of the priming manipulation.

Limitations Stated by Authors

- Although this result supports our hypothesis that social interaction behavior can be primed, the time-to-interruption distribution varied considerably from normality. Fully 21 of the 34 participants did not interrupt at all in the 10 min available to them, so that the time variable suffered from a severe ceiling effect.

Preview of Next Experiment

We designed Experiments 2 and 3 to test this hypothesis in the context of two distinct stereotypes: one for elderly people and one for African Americans.

Citations Referenced

- **Higgins & King (1981)** -- Higgins & King,1981
- **Neuberg (1988)** -- Neuberg,1988
- **Herr (1986)** -- Herr,1986
- **Bargh & Thein (1985)** -- Bargh & Thein,1985
- **Higgins (1989)** -- Higgins,1989
- **Newman & Uleman (1989)** -- Newman & Uleman,1989
- **Srull & Wyer (1979)** -- Srull & Wyer,1979
- **Brewer (1988)** -- Brewer,1988

- **Devine (1989)** -- Devine,1989
- **Fiske & Neuberg (1990)** -- Fiske & Neuberg,1990
- **Hayes-Roth (1977)** -- Hayes-Roth,1977
- **Bargh & Pietromonaco (1982)** -- Bargh & Pietromonaco,1982
- **Bargh et al. (1986)** -- Bargh et al.,1986
- **Srull & Wyer (1979)** -- Srull & Wyer,1979

Experiment 2a and 2b -- Introduction

Not available for this paper.

Experiment 2a and 2b -- Methods

Section: exp_2a_and_2b

Participants

- **N:** 60 (30 in Experiment 2a, 30 in Experiment 2b)
- **Recruitment:** New York University undergraduates
- **Testing context:** individual, lab, session length not reported

Design

Design statement: 2 (prime condition: elderly vs neutral) x 2 (experiment: 2a vs 2b) between-subjects design, with prime condition between participants in each experiment.

Predictors:

- **prime condition**
- Levels: elderly prime, neutral prime
- Manipulation: between-subjects
- Factor type: nominal

Counterbalancing: not reported

Outcome Measures

- **walking time:** time in seconds to walk 9.75 m down corridor measured with a hidden stopwatch

Materials

- **Description:** scrambled-sentence task with 30 sets of five word combinations; participants write a sentence using four of the five words
- **Source:** constructed by authors; prime words obtained from previous research (Brewer, Dull, & Lui, 1981; Harris & Associates, 1975; McTavish, 1971; Perdue & Gurtman, 1990)
- **Stimuli presented per participant:** 30 sets per participant
- **Additional details:**
- self-paced
- written sentences
- elderly prime version contained stereotype-relevant words

Procedure

Ordered steps:

1. Participant arrives individually and is told the study investigates language proficiency
2. Participant receives scrambled-sentence task instructions and completes 30 self-paced sets
3. Experimenter leaves the room while participant works
4. Participant notifies experimenter when finished; experimenter re-enters and partially debriefs
5. A confederate records walking time down a 9.75 m corridor using a hidden stopwatch
6. Experimenter meets participant near elevator and gives complete debriefing

- **Apparatus:** scrambled-sentence task sheets; hidden stopwatch; corridor of known length (9.75 m)
- **Experimenter presence:** present initially, leaves during task, returns for debriefing
- **Presentation format:** not reported
- **Stimulus timing:** self-paced

Preregistration and Analysis Plan

- **Preregistration:** not reported

Author-Provided Rationale for Methodological Choices

- to study the effect of activation of the elderly stereotype on behavior

Missing or Unclear

- no power analysis reported
- compensation not specified
- exclusion criteria not stated
- detailed demographic information not provided
- per-cell sample size not explicitly reported
- counterbalancing scheme not described
- session duration not reported
- stimulus timing details beyond self-paced not given

Citations Referenced in This Section

- **Brewer, Dull, & Lui (1981)** -- source of elderly prime words
- **Harris & Associates (1975)** -- source of elderly prime words
- **McTavish (1971)** -- source of elderly prime words
- **Perdue & Gurtman (1990)** -- source of elderly prime words

Quoted Evidence from Source

> Thirty male and female New York University undergraduates ... participated in Experiment 2a, and a different sample of 30 participated in Experiment 2b.

> Participants were randomly assigned to either an elderly prime condition or a neutral prime condition.

> Using a hidden stopwatch ... recorded the amount of time in seconds that the participant spent walking a length of the corridor ... 9.75 m away.

Experiment 2a and 2b -- Results and Discussion

Section: exp_2a_and_2b

Statistical Approach

A t test was computed to ascertain the effect of the priming manipulation on walking speed. A multivariate analysis of variance (MANOVA) was used to analyze the effect of priming condition on mood and arousal.

Key Findings

Participants in the elderly priming condition walked slower than those in the neutral condition in Experiment 2a

- **Test:** t test
- **Effect type:** main effect
- **Variables in model:** priming condition, walking speed

Verbatim statistical report:

> $t(28) = 2.86, p < .01$

Verbatim descriptives:

> Participants in the elderly priming condition ($M = 8.28$ s) had a slower walking speed compared to participants in the neutral priming condition ($M = 7.30$ s)

Participants in the elderly priming condition walked slower than those in the neutral condition in Experiment 2b

- **Test:** t test
- **Effect type:** main effect
- **Variables in model:** priming condition, walking speed

Verbatim statistical report:

> $t(28) = 2.16, p < .05$

Verbatim descriptives:

> Participants in the elderly priming condition ($M = 8.20$ s) had a slower walking speed compared to participants in the neutral priming condition ($M = 7.23$ s)

No significant main effect of priming condition on mood or arousal

- **Test:** MANOVA
- **Effect type:** main effect

- **Variables in model:** priming condition, emotion scales, arousal scales

Verbatim statistical report:

> $F(1,31) < 1$

Verbatim descriptives:

> Participants in the elderly priming condition were in a more positive mood ($M = 1.7$) than control participants ($M = 0.3$) and were more aroused or energetic as well ($Ms = -0.5$ and -1.2, respectively)

No significant interaction between priming condition and mood or arousal

- **Test:** MANOVA
- **Effect type:** interaction
- **Variables in model:** priming condition, emotion scales, arousal scales

Verbatim statistical report:

> $F(1,31) < 1$

Authors' Interpretations

stereotype activation influences behavior

> The results of the present experiments suggest that exposing individuals to a series of words linked to a particular stereotype influences behavior nonconsciously. How the activated stereotype influences behavior depends on the content of the activated stereotype itself, not the stimulus words actually presented.

nonconscious effect of priming

> Thus, it appears safe to conclude that the effect of the elderly priming manipulation on walking speed occurred nonconsciously.

mood does not mediate the effect

> The alternative explanation for our findings in terms of a mediating effect of mood caused by the elderly stereotype primes appears untenable.

Authors' Stated Links to Their Own Predictions

priming effect on walking speed

> Participants in the elderly priming condition ($M = 8.28$ s) had a slower walking speed compared to participants in the neutral priming condition ($M = 7.30$ s), $t(28) = 2.86$, $p < .01$, as predicted.

Alternative Interpretations Discussed by Authors

- One alternative explanation that can be offered for the effect of the elderly-stereotype-related stimuli on walking speed is that, in general, the words relating to the elderly stereotype are more likely than control words to induce in participants a sad mood, which might then be the reason they walked more slowly.

Missing or Unclear (Non-Statistical)

- No information on reliability estimates for the measures used
- No information on the exact coding procedures for the scrambled-sentence task

Citations Referenced

- **Bargh (1992)** -- review in Bargh, 1992
- **Lombardi, Higgins, & Bargh (1987)** -- Lombardi, Higgins, & Bargh, 1987
- **Strack & Hannover (1996)** -- Strack & Hannover, 1996
- **Page (1969)** -- Page, 1969
- **Salovey & Birnbaum (1989)** -- Salovey & Birnbaum, 1989

Experiment 3 -- Introduction

Section: exp_3

Rationale for This Experiment

To test whether the automatic activation effects observed for the elderly stereotype also apply to a different stereotype (African American), thereby assessing the generality of the model.

Relation to Prior Experiments

Builds on Experiment 2, which showed automatic activation of the elderly stereotype produced direct, nonconscious behavioral effects; Experiment 3 extends this to a new stereotype.

Specific Predictions

hostile behavior tendency

> according to the present model, this automatic activation should also produce a tendency toward hostile behavior in the perceiver.

- Citations: Devine (1989)

Theoretical Refinements (relative to paper-level intro)

generality to different stereotype

> Experiment 3 was intended to assess the generality of these results to an entirely different stereotype--that for African Americans.

New Elements Introduced

- Priming stimuli presented subliminally, eliminating conscious, strategic processes
- Different target stereotype (African American) instead of elderly

Constraints Addressed

- Rules out experimenter demand or conscious strategic processes as alternative explanations

Missing or Unclear

- Specific details about the behavioral measure used to assess hostile behavior are not provided

Citations Referenced

- **Devine (1989)** -- automatic activation of the African American stereotype influencing perceptions of hostility

Experiment 3 -- Methods

Section: exp_3

Participants

- **N:** 41
- **Recruitment:** New York University
- **Compensation:** partial course credit
- **Testing context:** individual, lab, session length not reported

Design

Design statement: 2 (priming: African American vs. Caucasian) between-subjects design.

Predictors:

- **priming**
- Levels: African American face, Caucasian face
- Manipulation: between-subjects
- Factor type: nominal

Counterbalancing: not reported

Outcome Measures

- **hostility rating:** 5-point unipolar scale (0-10) rated by two blind coders from videotaped facial expressions, averaged across coders

Materials

- **Description:** Black-and-white photographs of young African American or Caucasian male faces, two pattern masks, and target pictures composed of 4-20 colored circles on a gray background
- **Source:** in-house stimuli, pretested for masking effectiveness
- **Stimuli presented per participant:** 130 face presentations per participant
- **Additional details:**
- face presentation 13-26 ms (monitor 76 Hz)
- target picture displayed for 3 s

Procedure

Ordered steps:

1. Participant brought individually to experimental room

2. Given instructions about odd-even judgment task
3. Practice trials administered
4. Experimenter leaves room
5. Subliminal face prime presented before each trial (African American or Caucasian, randomly assigned)
6. Target picture shown, participant makes odd/even response
7. After trial 130, computer displays error message and instructs participant to get experimenter
8. Experimenter returns, announces need to repeat task, then rescinds
9. Videotaped facial reactions recorded throughout
10. Two blind coders rate hostility on 5-point scale
11. Participant completes two questionnaires (Racial Ambivalence Scale, Modern Racism Scale)
12. Experimenter rates irritability, hostility, anger, uncooperativeness on 10-point scales
13. Suspicion probe and debriefing

- **Apparatus:** Gateway 486 computer with VGA monitor, Visual Basic program, hidden video camera
- **Experimenter presence:** present initially, leaves during task, returns after error
- **Presentation format:** trials presented sequentially; participant randomly assigned to priming condition by computer
- **Stimulus timing:** face 13-26 ms, masks similar duration, target 3 s; trial timing bounded by hardware

Preregistration and Analysis Plan

- **Preregistration:** not reported

Author-Provided Rationale for Methodological Choices

- Pretesting showed masking procedure was effective and participants were unaware of face photographs
- Subliminal presentation (13-26 ms) used to ensure unconscious priming

Missing or Unclear

- Power analysis not reported
- Exclusion criteria not stated
- Detailed demographic information (age, gender, ethnicity beyond non-African-American) not provided
- Exact number of stimulus items in the pool not given
- Counterbalancing or randomization details beyond computer assignment not described
- Session length / total testing time not reported
- Specific ISI between face and mask not reported

Citations Referenced in This Section

- **Katz & Hass (1988)** -- source of the Racial Ambivalence Scale used after the task
- **McConahay (1986)** -- source of the Modern Racism Scale used after the task

Quoted Evidence from Source

> Participants were 41 non-African-American undergraduate students from New York University who participated in the experiment to partially fulfill course credit.

> The hypothesis was that participants presented subliminally with African American faces would react with greater hostility to the computer error, compared to participants primed with Caucasian faces.

> Two coders who were blind to experimental conditions and hypotheses rated all the videotaped facial expressions ... on a 5-point unipolar scale of hostility ranging from 0 (not at all hostile) to 10 (extremely hostile).

Experiment 3 -- Results and Discussion

Section: exp_3

Statistical Approach

We conducted a MANOVA using an average of the experimenter's hostility ratings across the four trait scales and an average of the two videotape coders' hostility ratings as dependent measures, and priming condition as the independent measure.

Key Findings

Participants primed with African American faces showed more hostility than those primed with Caucasian faces

- **Test:** MANOVA
- **Effect type:** main effect
- **Variables in model:** priming condition, hostility ratings (experimenter and coders)

Verbatim statistical report:

> $F(1,39) = 6.95$, $p < .05$

Verbatim descriptives:

> Participants primed with photographs of African American faces behaved in a more hostile fashion ($M = 2.79$) compared to participants primed with Caucasian faces ($M = 2.13$)

No significant difference in hostility ratings between experimenter and blind coders

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** rating source (experimenter vs. blind coders), hostility ratings

Verbatim statistical report:

> $F(1,39) < 1$

No interaction between rating source and priming condition on hostility ratings

- **Test:** ANOVA
- **Effect type:** interaction
- **Variables in model:** rating source (experimenter vs. blind coders), priming condition, hostility ratings

Verbatim statistical report:

> $F(1,39) < 1$

No significant correlation between hostility and racism measures in either priming condition

- **Test:** correlation

- **Effect type:** correlation
- **Variables in model:** hostility measure, racism measures (Racial Ambivalence Scale and Modern Racism Scale)

Verbatim statistical report:

> all rs < .25, ps > .35

Authors' Interpretations

Hostility behavior difference

> participants primed with photographs of African American faces behaved in a more hostile fashion (M = 2.79) compared to participants primed with Caucasian faces (M = 2.13)

No moderation by racism

> participants who were low in racist attitudes toward African Americans were just as likely to behave in a hostile manner as participants who were high in racist attitudes, regardless of priming condition

Authors' Stated Links to Their Own Predictions

Correspondence to Devine (1989)

> This finding corresponds to that of Devine (1989, Experiment 2), who found that the automatic effect of the African American stereotype on social perception did not vary as a function of level of consciously expressed prejudice, as measured by the Modern Racism Scale

- Citations: Devine (1989)

Correspondence to Fazio et al. (1995)

> More recently, Fazio, Jackson, Dunton, and Williams (1995) also found no moderation by Modern Racism scores of their obtained effect of racial attitudes on behavior

- Citations: Fazio, Jackson, Dunton, and Williams (1995)

Citations Referenced

- **Devine (1989)** -- Experiment 2 on automatic effect of African American stereotype
- **Fazio, Jackson, Dunton, and Williams (1995)** -- study on racial attitudes on behavior

- **Lepore & Brown (1996)** -- study on moderation by Modern Racism scores

General Discussion

Section: paper-level

Summary of Findings

Trait or stereotype activation in one context produced consistent behavior in a later unrelated context, without participants' awareness, and these effects were stronger for behavior than for impression measures.

Theoretical Conclusions

situationally linked behavior

> One explanation for the present results is in terms of situationally linked behavior; that is, that a person has behavioral responses (e.g., assertiveness, anger, patience) that are associatively linked to particular situations (e.g., being made to wait by another person, losing one's work because of another's mistake).

automatic perception-behavior link

> another theoretical perspective consistent with the present findings is that of an automatic perception-behavior link (e.g., Berkowitz,1984; Carver et al.,1983).

- Citations: Berkowitz (1984); Carver et al. (1983)

behavioral schema model

> Thus, our results are in harmony with those of Carver et al. (1983), who advanced a "behavioral schema" model as an explanation for modeling effects.

- Citations: Carver et al. (1983)

conditions for automatic behavior

> automatic social behavior may occur only if the behavioral representation that is activated is already associated with that situation by the individual.

control of automatic behavior

> Control over automatic influences requires three things: (a) awareness of the influence or at least the possibility of the influence, (b) motivation to exert the control, and (c) enough attentional capacity (or lack of distractions) at the time to engage in the control process (see Bargh,1989; Wegner,1994).

- Citations: Bargh (1989); Wegner (1994)

automatic stereotype-based self-fulfilling prophecy

> there may be an automatic, nonconscious basis for self-fulfilling prophecy effects (e.g., Snyder, Tanke, & Berscheid,1977).

- Citations: Snyder, Tanke, & Berscheid (1977)

automatic empathic reactions

> the perception-behavior link may be an important ingredient in the "glue" that binds two (or more) interaction partners, keeps them on the same wavelength, and helps to bring each partner a sense of validation by others of their experience.

Cross-Experiment Integration

behavioral priming across experiments

> Across three experiments, the activation of a trait construct or a stereotype in one context resulted in behavior consistent with it in a subsequent unrelated context.

Results Authors State Support Their Theory

situationally linked behavior support

> The present results show that these behavioral responses become automatically linked to representations of social situations just as previous research has found perceptual trait constructs, stereotypes, and attitudes to become automatically activated.

Results Authors State Challenge a Theory

challenge to mediation model

> In Experiment 1, impressions of the experimenter were not affected by the priming manipulation, whereas behavior strongly was.

Alternative Accounts Addressed by Authors

mediational model of behavior *(ad-hoc alternative)*

- **Description:** behavior is mediated by social perceptions and judgments

- **How authors addressed it:** acknowledged but argued that behavior can occur without mediation
- **Citations:** Bargh (in press); Bem (1972)

Implications for the Field

strength of behavioral priming vs. impression priming

> obtaining priming differences on these behavioral measures was considerably easier than in our laboratory's previous research on social judgment measures.

limits of mediation models

> behaviors can be triggered directly without being mediated by impressions or judgments of the person with whom one is interacting.

Applied Implications Stated by Authors

- Automatic priming may underlie self-fulfilling prophecy effects in stereotype contexts.
- Automatic perception-behavior links could influence empathic reactions and social cohesion.

Limitations Stated by Authors

- The limited range of behavioral options may account for the strength of the findings.
- Cannot directly compare strength of priming effects on behavior versus judgments due to many differences between experimental situations.

Future Directions Proposed by Authors

- Investigate the boundary conditions under which automatic behavior can be controlled or overridden.
- Examine how individual differences in behavioral repertoires affect automatic priming effects.

Citations Referenced

- **Lewin (1943)** --
- **Mischel (1973)** --
- **Berkowitz (1984)** --

- **Higgins (1987)** --
- **Carver et al. (1983)** --
- **Devine (1989)** --
- **Srull & Wyer (1979)** --
- **Prinz (1990)** --
- **Kihlstrom, Barnhardt, & Tataryn (1992)** --
- **Moore (1982)** --
- **Bargh (1989)** --
- **Wegner (1994)** --
- **Snyder, Tanke, & Berscheid (1977)** --
- **Dodge & Crick (1990)** --
- **Nisbett & Wilson (1977)** --

Critique

Paper: bargh-et-al-2015-automaticity-of-social-behavior--direct-effects-of-trait-construct-and-stereotype-activation-on-a

Overall Assessment

The paper suffers from several high-severity methodological shortcomings, most notably the absence of any reported power analysis, which casts doubt on the adequacy of its sample sizes. While the theoretical framework is articulated, the operationalization of key constructs and the alignment between theory and measured outcomes are often tenuous, and alternative mediated pathways are insufficiently addressed. Moreover, the statistical reporting lacks essential details such as effect sizes, confidence intervals, and appropriate corrections for multiple comparisons, further weakening the evidentiary support for the claimed direct, nonconscious effects. Consequently, the study's rigor and evidential robustness are limited, reducing its standing as a reliable contribution to the literature.

1. Design-Theory Alignment

- **The interruption latency outcome in Experiment 1 operationalizes a general willingness to intervene rather than a direct behavioral manifestation of the primed rudeness construct, potentially conflating assertiveness with the targeted trait.**
- Severity: **moderate**

- Justification: The theory posits that priming a rudeness trait should automatically trigger rude behavior. Measuring how quickly participants interrupt an ongoing conversation captures a willingness to act, but it also reflects general assertiveness or impatience, which are not uniquely tied to rudeness. Without a more specific behavioral indicator (e.g., overtly rude remarks), the outcome may not cleanly map onto the theoretical construct, limiting the ability to attribute the effect to automatic trait activation.
- Evidence:
- `methods/exp_1` -- field `outcome_measures`: Outcome measured as time from entering hallway until participant spoke to the experimenter (interruption latency).
- `results_and_discussion/exp_1` -- field `key_findings`: Finding that rude-prime participants interrupted faster, interpreted as a direct effect of trait activation.
- `introduction` -- field `constructs_of_interest`: Outcome defined broadly as "behavioral responses" presumed to be automatically activated.

2. Evidence-Theory Alignment

- **The authors claim a direct, preconscious effect of trait priming on interruption behavior in Experiment 1, but the discussion does not adequately rule out alternative perceptual explanations for the faster interruptions.**
- Severity: **moderate**
- Justification: The theoretical claim is that trait concepts become automatically activated and directly drive behavior. Because the presented results leave a plausible perceptual mediation untested, the conclusion overreaches the empirical evidence. A more cautious interpretation would be warranted.

Evidence:

- `results_and_discussion/exp_1` -- field `authors_interpretations`: Authors interpret the faster interruptions as a direct, preconscious effect while also noting an alternative explanation that participants may have perceived the experimenter differently based on the prime (e.g., rude-primed participants perceiving rudeness).
- `results_and_discussion/exp_1` -- field `alternative_interpretations_discussed`: They discuss a perception-based alternative but do not provide additional data (e.g., manipulation checks) to disconfirm it, yet they still present the behavior as evidence for automatic activation.

In Experiment 2 the authors assert a nonconscious effect of the elderly stereotype on walking speed, yet they report a significant mood difference (more positive mood

in the elderly prime) while simultaneously claiming mood does not mediate the behavioral effect.

- Severity: **moderate**
- Justification: The claim that the behavioral effect is nonconscious and mood-independent conflicts with the reported mood data, suggesting an overstatement of the evidence supporting the theory that stereotype activation directly drives behavior without affective mediation.

Evidence:

- `results_and_discussion/exp_2a_and_2b` -- field `key_findings`: Finding: No significant main effect of priming on mood or arousal (MANOVA F<1) but descriptives show participants in the elderly prime were more positive (M=1.7 vs. 0.3).
- `results_and_discussion/exp_2a_and_2b` -- field `authors_interpretations`: Authors conclude the effect on walking speed occurred nonconsciously and that mood mediation is untenable, despite the reported mood difference.

Experiment 3 generalizes the automatic activation model to hostile behavior toward an African-American prime, but the behavioral measure (hostility ratings after a computer error) may not directly reflect the hypothesized stereotype-driven hostility, and the authors do not address this potential construct mismatch.

- Severity: **minor**
- Justification: The link between the priming manipulation and the observed hostility rating is indirect; without a clear justification that the error-induced hostility captures the theorized stereotype-driven hostile response, the claim stretches the evidence beyond what was directly measured.
- Evidence:
- `exp_introduction/exp_3` -- field `specific_predictions`: Prediction: automatic activation should produce a tendency toward hostile behavior.
- `results_and_discussion/exp_3` -- field `key_findings`: Finding: Participants primed with African-American faces showed higher hostility ratings (M=2.79 vs. 2.13) after a computer error.
- `exp_introduction/exp_3` -- field `missing_or_unclear`: Specific details about the behavioral measure used to assess hostile behavior are not provided.

4. Alternative Accounts

- **The authors dismiss the mediational model of behavior (behavior mediated by social perceptions) without providing a thorough theoretical or empirical justification, despite extensive literature supporting mediated pathways.**
- Severity: **moderate**
- Justification: While the paper raises the mediational account as a competing theory, it does not present data (e.g., mediation analysis) or a robust theoretical argument to rule it out. Given the prominence of mediated models in social cognition, a more detailed evaluation is warranted to strengthen the claim of direct automatic behavior.
- Evidence:
- `general_discussion` -- field `alternative_accounts_addressed`: The mediational model is listed as an alternative account but is only briefly acknowledged and argued against without detailed evidence.
- `results_and_discussion/exp_1` -- field `authors_interpretations`: Authors claim a direct preconscious effect on behavior while noting an alternative explanation involving social perceptual effects, yet they do not empirically test the mediation hypothesis.

5. Statistical Completeness

- **The 2 × 2 between-subjects design (prime condition × experiment 2a/2b) is not fully analyzed; the main effect of experiment and the prime × experiment interaction are never reported.**
- Severity: **moderate**
- Justification: In a factorial ANOVA with two factors, a complete analysis should include the main effect of each factor and their interaction. The authors present only the priming main effect within each experiment and omit any test of whether the experiment (2a vs 2b) itself influences the outcome or interacts with priming. This omission leaves the reported statistical analysis incomplete relative to the stated design.
- Evidence:
- `methods/exp_2a_and_2b` -- field `design_statement`: 2 (prime condition: elderly vs neutral) x 2 (experiment: 2a vs 2b) between-subjects design
- `results_and_discussion/exp_2a_and_2b` -- field `key_findings`: Only separate t-tests for each experiment and a MANOVA on mood/arousal are reported; no statistics for a main effect of experiment or for the prime × experiment interaction are provided

6. Methodological Concerns

- **No power analysis was reported for any of the experiments, leaving the adequacy of sample sizes unclear.**
- Severity: **high**
- Justification: Without a priori power calculations, it is impossible to assess whether the per-condition Ns (e.g., 34 total in Exp 1, 30 per group in Exp 2, 41 total in Exp 3) provide sufficient statistical power to detect the hypothesized effects. This threatens the validity of the reported significant findings.

Evidence:

- `methods/exp_1` -- field `power_analysis`: power_analysis: null
- `methods/exp_2a_and_2b` -- field `power_analysis`: power_analysis: null
- `methods/exp_3` -- field `power_analysis`: power_analysis: null

Exclusion criteria were not specified for any experiment, making participant selection and data cleaning procedures ambiguous.

- Severity: **moderate**
- Justification: Clear exclusion rules are essential for reproducibility and for understanding potential biases introduced by post-hoc data removal. Their absence limits transparency.

Evidence:

- `methods/exp_1` -- field `exclusions`: exclusions: [] (no criteria stated)
- `methods/exp_2a_and_2b` -- field `exclusions`: exclusions: [] (no criteria stated)
- `methods/exp_3` -- field `exclusions`: exclusions: [] (no criteria stated)

Counterbalancing procedures are missing or insufficiently described for experiments that required it (Exp 2 and Exp 3).

- Severity: **moderate**
- Justification: Both experiments used between-subjects manipulations that could be confounded with order or stimulus presentation effects; without a described counterbalancing or randomization scheme, the internal validity of the priming manipulation is uncertain.

Evidence:

- `methods/exp_2a_and_2b` -- field `counterbalancing`: counterbalancing: "not reported"
- `methods/exp_3` -- field `counterbalancing`: counterbalancing: "not reported"

Reliability estimates for subjectively coded measures (e.g., hostility ratings in Exp 3 and interruption ratings in Exp 1) are not reported.

- Severity: **moderate**
- Justification: Inter-rater reliability is crucial when using human coders; without it, the measurement precision of the key dependent variables cannot be evaluated.

Evidence:

- `results_and_discussion/exp_3` -- field `missing_reliability`: No reliability estimates for the two blind coders' hostility ratings
- `results_and_discussion/exp_1` -- field `missing_reliability`: No reliability reported for experimenter politeness ratings or for the interruption latency coding

Preregistration status is not reported for any experiment, despite the detailed methodological approach.

- Severity: **minor**
- Justification: While not mandatory, preregistration enhances transparency and guards against analytic flexibility. Its omission reduces confidence in the pre-specified nature of hypotheses and analyses.

Evidence:

- `methods/exp_1` -- field `preregistration_status`: preregistration_status: "not reported"
- `methods/exp_2a_and_2b` -- field `preregistration_status`: preregistration_status: "not reported"
- `methods/exp_3` -- field `preregistration_status`: preregistration_status: "not reported"

Compensation details are missing for Exp 1 and Exp 2, limiting assessment of potential participant motivation effects.

- Severity: **minor**
- Justification: Knowing whether participants received course credit, payment, or no compensation helps evaluate possible demand characteristics or selection biases.
- Evidence:
- `methods/exp_1` -- field `compensation`: compensation: null
- `methods/exp_2a_and_2b` -- field `compensation`: compensation: null

7. Statistical Robustness

- **Effect sizes and confidence intervals are not reported for any of the primary hypothesis tests, limiting assessment of the practical significance of the findings.**
- Severity: **moderate**
- Justification: Without effect size metrics or confidence intervals, readers cannot gauge the magnitude or reliability of the reported effects. This omission hampers evaluation of whether statistically significant results correspond to meaningful behavioral changes.

Evidence:

- `results_and_discussion/exp_1` -- field `verbatim_statistical_report`: $F(2,33) = 5.76$, $p = .008$; t tests reported as both ps < .04 without any η^2, d, or CI.
- `results_and_discussion/exp_2a_and_2b` -- field `verbatim_statistical_report`: $t(28) = 2.86$, $p < .01$ and $t(28) = 2.16$, $p < .05$ with no effect size or confidence interval.
- `results_and_discussion/exp_3` -- field `verbatim_statistical_report`: $F (1,39) = 6.95$, $p < .05$ reported without effect size or CI.

Planned pairwise t-tests in Experiment 1 are presented without any correction for multiple comparisons, raising the risk of inflated Type I error.

- Severity: **moderate**
- Justification: Conducting multiple planned comparisons increases the familywise error rate. Reporting unadjusted p-values may overstate the evidence for differences between the rude prime and each other condition.

Evidence:

- `results_and_discussion/exp_1` -- field `verbatim_statistical_report`: Both ps < .04 for simple t tests following a significant ANOVA, but no mention of Bonferroni, Holm, or other correction.

Non-significant mood effects in Experiment 2 are described in narrative terms (e.g., "more positive mood") despite $F(1,31) < 1$, which could mislead readers about the presence of an effect.

- Severity: **minor**
- Justification: Describing directionality of non-significant results may give the impression of an effect where statistical evidence does not support it, potentially overstating the findings.

Evidence:

- `results_and_discussion/exp_2a_and_2b` -- field `verbatim_descriptives`: Participants in the elderly priming condition were described as "more positive mood (M =

1.7) than control participants (M = 0.3)" even though the MANOVA reported F(1,31) < 1 (non-significant).

The paper does not report any adjustments for the multiple experiments (three experiments) when interpreting overall significance, which may lead to cumulative Type I error inflation across the study series.

- Severity: **minor**
- Justification: When several independent experiments each report significant p-values, the overall false-positive rate across the set of tests is higher than the nominal α level unless a correction (e.g., Bonferroni across experiments) is applied.
- Evidence:
- `general_discussion` -- field `cross_experiment_integration`: Authors claim "Across three experiments, the activation... resulted in behavior consistent with it" without acknowledging the need for a correction for multiple independent tests.